In sum, the series of events is as follows:

1. The user is presented with a page that consists of a welcome message and a textbox.
2. The user types a question and presses Enter.
3. The web UI takes the user's question and submits it to the backend service.
4. The backend accepts the query and processes it.
5. The backend submits the query along with some context to the suitable AI model that can understand as well as generate natural language or code.
6. The AI model responds with a query response.
7. The backend service processes the response and verifies its contents.
8. The backend sends the cleaned response back to the web UI.
9. The web UI updates the list of conversation messages.
10. The user can submit follow-up questions.

2.2.2 Project and technology requirements

Given the sequence of events, we can now list our initial requirements for building Astra AI as a simple, user-friendly, and effective chat app. In the first iteration, we want to prioritize a foundational experience. As I introduce concepts throughout this book, we will be enhancing Astra's AI functionality to suit our growing needs:

- *A single page*—The user interface should consist of a single webpage.
- *Welcome message*—A clear and welcoming message should greet the user, introducing Astra AI and its capabilities.
- *Text input box*—A prominent text input box should be readily available for users to type their questions in natural language.
- *Conversation history*—A dedicated area should display the entire conversation history between the user and Astra AI. This allows users to refer to previous interactions and provides context for follow-up questions.
- *Scrolling functionality*—The conversation history area should be scrollable, enabling users to revisit earlier parts of the conversation if needed.
- *Visual design*—A simple interface is preferred for this initial iteration.

To build the Astra AI's initial iteration, we'll use the following technologies. Don't worry if you're unfamiliar with some of the tools mentioned. The example projects are preconfigured to run without any changes on your part. Consequently, you can follow along even if you're new to these tools:

- *Frontend*—React.js
- *Backend*—Node.js
- *CSS framework*—Tailwind.css
- *Build tool*—Vite